

**On Faultization:
Rubik.
What Perturbation Reveals About the Scale of Competence
in an Exactly Analyzable Collective**

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Abstract

Fault tolerance asks whether a designed function survives damage. We develop the reverse question under the proposed name faultization: decompose a centralized algorithm into locally competent parts, damage them deliberately, and use the damage to locate where competence resides and at what scale a goal-directed description becomes predictive. The pocket cube provides a substrate on which every alternative explanation can be enumerated: 3,674,160 reachable states, an exact distance oracle, and exhaustively computable controller dynamics. Seven corner agents, each knowing its correct pose and predicting every move's effect on itself, collectively solve 61 of 3,674,160 states (0.0017 percent), while 758,286 states (20.6 percent) are local optima in which no agent consents to move; in 72 percent of states no agent is satisfied, and the group's parity makes exactly 6 correct corners impossible. Lateral moves create 145,953 attractors and leave the basin at 61, as does one remembered action; the best of 48 random weightings reaches 136. Deleting 3 of the 7 voters raises the mean basin to 91.8, with patterns ranging from 19 to 260. Action noise at its optimal rate of 0.34 raises success within 200 steps 119-fold, to 0.0020. Deleting one face's moves makes recovery impossible from 99.2 percent of states, separating incompetence from impossibility. Under strict arbitration, conflicting targets define a mixed potential with no cycles, and both targets survive as equilibria among roughly 729,000 to 778,000 equilibria. Retargeting leaves the basin at 61 by group conjugacy. Together these deflationary results form a catalogue of cheaper explanations that claims of emergent collective agency must exclude.

Introduction

Fault tolerance asks whether a specified service survived damage. The question has decades of good answers, and every answer treats the fault as something to be withstood. We develop the complementary question of what the damage makes visible. When a centralized algorithm is decomposed into locally competent parts, and the parts are deleted, frozen, miscalibrated, given incompatible goals or denied information, the informative outcome is what the perturbation reveals: where the competence resides, what the arbitration layer contributed, which attractors the dynamics contained, and at what scale a goal-directed description begins to predict. We call the method faultization; the term is proposed here and does not belong to an established field.

A faultization experiment is specified by a decomposition into agents, an observability assignment, an actuation model, a communication budget, a target representation, a fault model and an arbitration rule. If any of these is left unspecified, claims of emergent competence cannot be identified, because the competence may reside in the scheduler, the aggregation rule, the target detector or the training procedure. The method also requires the dependability tradition's separation of fault, error and failure (Avizienis, Laprie, Randell and Landwehr, 2004). Noise is transient corruption of a signal. A fault is a defect capable of producing an error. Damage is a persistent structural alteration. Local incompetence is an intact policy failing its own objective. Distributed partial competence, the condition of interest, consists of components that each solve a real subproblem while none holds the global policy. These are 5 distinct interventions with different consequences, and pooling them as "errors" produces conclusions stronger than the evidence.

The self-sorting array experiments

The direct precedent is the self-sorting arrays of the Levin group, which supply both the idea and a cautionary example (Zhang, Goldstein and Levin, 2025). Bubble, Insertion and Selection sort were reimplemented as collections of cell agents, one per array entry, each carrying a position, a fixed value and a fixed local policy and running in its own thread. Faults were frozen cells of two grades: passive cells that cannot initiate a move but can be displaced, and stuck cells that cannot move at all. The distributed implementations often retained more final order than their centralized counterparts, the ranking of algorithms reversed between the two fault grades, mixtures of policies still sorted and transiently clustered by policy, and populations given opposite sorting directions settled into stable macroscopic compromises. These findings are real and nontrivial.

The implementation also kept a main thread that activates, monitors and terminates the agents, a global lock that serializes every swap, and a coin flip before each lock acquisition, which together amount to a random fair scheduler and a centralized arbitration resource. Only the Bubble cells are strictly local: the Insertion cells read every cell to their left and a global property of the cells to their right, and the Selection cells carry explicit positional targets. Decentralized execution and local information are different properties, and only the first was fully delivered. The most quoted statistic, a delayed-gratification measure built from temporary decreases in sortedness, rose for Bubble cells from 0.24 with no frozen cells to 0.37 with 3. The authors note that a random walker also moves away from its target, call for barrier-conditioned controls, and state that the implementation can be classified with random sorting algorithms and involves nothing beyond its rules. A neutral name for the observed effect is barrier-correlated nonmonotonic progress, and the published record does not yet separate it from stochastic wandering under an asynchronous scheduler. That separation requires a substrate on which every alternative explanation can be enumerated.

Self-stabilization and related theory

Recovery without a dedicated fault handler has been formalized since 1974. A protocol is self-stabilizing when every execution from every initial state eventually reaches a legitimate state and legitimate states lead only to legitimate states, the properties of convergence and closure formalized by Dijkstra for locally interacting processes on a token ring (Dijkstra, 1974; Schneider, 1993). Randomization can produce convergence where deterministic symmetry prevents it, and fairness assumptions do real work there (Herman, 1990). Fault containment added the requirement that recovery cost scale with the fault and not with the system (Ghosh, Gupta, Herman and Pemmaraju, 1996). Byzantine analysis established how to state what a collective guarantees against components that lie (Lamport, Shostak and Pease, 1982). Work on locality proved that what bounded-radius agents can compute at all depends on anonymity, identifiers and scheduling, before any question of efficiency (Naor and Stockmeyer, 1995; Angluin, Aspnes, Diamadi, Fischer and Peralta, 2006). The oldest result shows that architecture can make a system more reliable than any of its parts (Von Neumann, 1956), and in biology strong robustness runs through degeneracy, structurally different components with overlapping functions, more than through duplication (Edelman and Gally, 2001).

A collective that regulates toward a target without a fault detector therefore demonstrates, by itself, a property formalized five decades ago, and a stronger claim must meet a higher standard. Apparent emergence often has a simpler source: neural cellular automata that regrow damaged patterns were trained on damaged intermediates, so their repair was implemented, though not legibly (Mordvintsev, Randazzo, Niklasson and Levin, 2020). Evolved cellular automata performing global computation are an earlier example (Mitchell, Crutchfield and Hrabner, 1994). Annealing and stochastic resonance, the two ordinary explanations for beneficial noise, complete the list of deflationary accounts an emergence claim must exclude (Kirkpatrick, Gelatt and Vecchi, 1983; Gammaitoni, Hänggi, Jung and Marchesoni, 1998).

The pocket cube as a test substrate

The fixed-corner pocket cube is the $2 \times 2 \times 2$ cube with one corner held still: 7 moving corners and a group of order $7! \times 3^6 = 3,674,160$, generated by the 9 face turns of the 3 faces that avoid the fixed corner. In an absolute frame the count is 88,179,840; the quotient by the 24 rigid rotations is the appropriate object for a controller that should not need an external reference frame. The full $3 \times 3 \times 3$ cube has 43,252,003,274,489,856,000 states and a diameter of 20 in the half-turn metric (Rokicki, Kociemba, Davidson and Dethridge, 2014), and its optimal solvers use pattern databases, a form of centralized control with global information (Korf, 1997; Joyner, 2008). The pocket cube fits entirely in memory. Our implementation reproduces the published distance shells exactly, 1, 9, 54, 321, 1,847, 9,992, 50,136, 227,536, 870,072, 1,887,748, 623,800 and 2,644, summing to 3,674,160 with diameter 11, and this reproduction calibrates every number below (Figure 1, left).

Four elementary facts set the rules of evidence. First, a fixed sequence of face turns is a permutation of the state set, so no open-loop sequence can send every configuration to solved, and feedback is necessary. Second, minimal feedback suffices for the weakest form of success: a random walk that checks after each move whether the target is reached, and stops there, converges from every state with probability 1. Universal convergence alone is therefore weak evidence of competence, and the corresponding control is priced below. Third, on any finite connected graph a spanning tree rooted at the target defines a feedback policy whose every step decreases tree distance, so a globally informed controller with the target as its unique attractor always exists; any failure under locality is informational. Fourth, reversibility is no obstruction: each generator is a bijection, but a state-dependent controller applies different generators in different states, so the closed-loop map can be many-to-one, with fixed points, basins and cycles. Because the solved state cannot be absorbing if a face must turn at every step, the controller includes a no-op, and homeostasis means remaining at the target as well as reaching it.

Composition of local correctness

In the first experiment the agents receive everything a generous designer could justify. Each of the 7 moving corners knows its target pose and its current pose, and predicts without error whether any candidate move would place it at its target. Each corner votes for the moves that would. An arbiter blind to the target and the state sums the votes, executes the first move in a fixed order that

strictly increases the number of satisfied agents, and stops when no move does. This is distributed partial competence in its cleanest form: every agent is locally infallible and none holds the global policy.

We computed the closed loop for all 3,674,160 states. The solved basin contains 61 states, 0.0017 percent. There are 758,286 non-target fixed points, 20.6 percent of the space, at which every move strictly decreases the number of satisfied agents and the collective does not act. The score landscape explains the failure: in 72 percent of all states no agent is satisfied, so the signal the collective climbs is absent almost everywhere, and where it exists the climb is short. The group adds a further constraint. Its permutation and orientation constraints make a state with exactly 6 of 7 corners correct impossible, so a score of 6 never occurs, and the final step to success always requires 2 agents to become correct together, which a controller built on single-agent consent cannot anticipate (Figure 1, right). Complete and error-free local correctness therefore does not compose into global competence, for a structural reason: every face turn moves 4 corners at once, and single-agent scores cannot express that coupling.

Allowing lateral moves makes the dynamics richer without improving them. An arbiter that moves whenever no strict decrease is available converts the 758,286 dead ends into 145,953 attractors: 80,127 fixed points holding 1,780,448 states, 43,774 cycles of length 2 holding 1,205,629, 2,512 cycles of length 3 holding 104,887, and 19,540 cycles of length 4 holding 583,196. The solved basin remains 61. A minor change in arbitration reorganized a fifth of the space into 145,953 attractors without affecting competence, which shows that the arbiter is part of the system and must be examined as such.

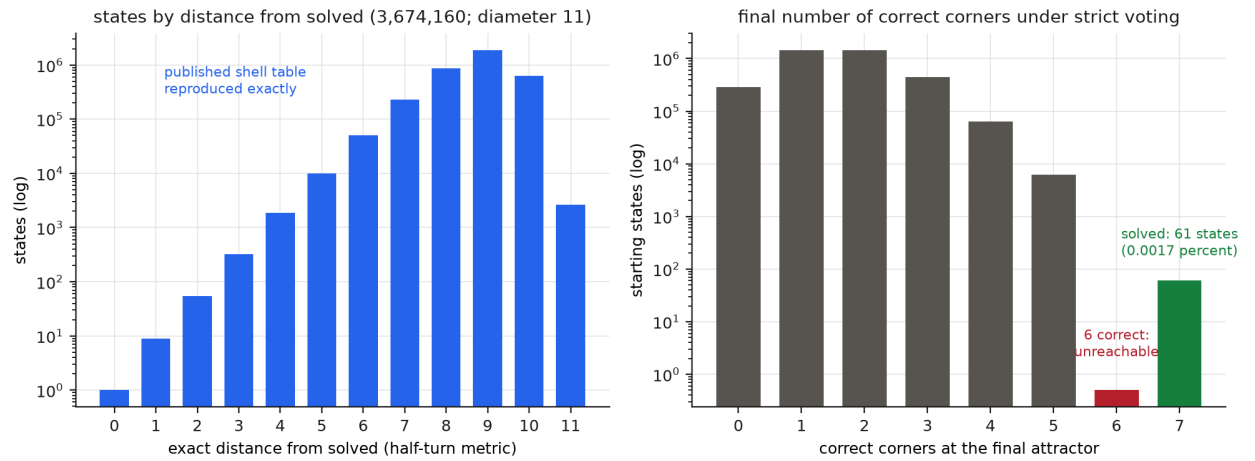


Figure 1: The pocket cube and the composition result. Left: the complete state space by exact distance from solved; the published shell table is reproduced exactly, diameter 11. Right: final number of correct corners for all 3,674,160 states under strict local-correctness voting; the solved basin is 61 states, and a final state with exactly 6 of 7 corners correct is impossible.

Deterministic and stochastic repairs

We evaluated three deterministic repairs. The first is memory: the collective remembers 1 action and may not undo it, the standard remedy for 2-cycles, computed exhaustively on the augmented space of 36,741,600 state-memory pairs. The solved basin is 61; the 2-cycles dissolve and trajectories reach other traps. The second is heterogeneity: the 7 agents' voting weights are drawn at random in 48 seeded draws. The best draw reaches a basin of 136, the worst 24 and the median 64, against 61 for homogeneous weights, so weights move the basin in both directions, as rearranged tie-breaks would, with no systematic gain. The third is the lateral-move variant above, which leaves the basin unchanged.

Randomness helps by a bounded amount. Mixing the deterministic controller with a uniform random face turn at rate ϵ , computed by exact iteration over the full state vector, gives success within 200 steps of 0.000017 at $\epsilon = 0$ and 0.000046 at $\epsilon = 1$. On the grid of rates examined, the largest value is 0.0019 at $\epsilon = 0.4$, and golden-section refinement places the optimum at $\epsilon = 0.34$ with success 0.0020 (Figure 2, left). The curve has the shape of an annealing curve, and its endpoints serve as controls. The right endpoint is the trivial universal solver, a random walk with a stop rule, which converges from every state with probability 1 but achieves 0.000046 within the horizon. The optimum exceeds it 43-fold, so the voting landscape carries information, and exceeds the deterministic controller 119-fold, so exploration contributes. The absolute success at the optimum is still 0.0020, and neither ratio indicates collective competence.

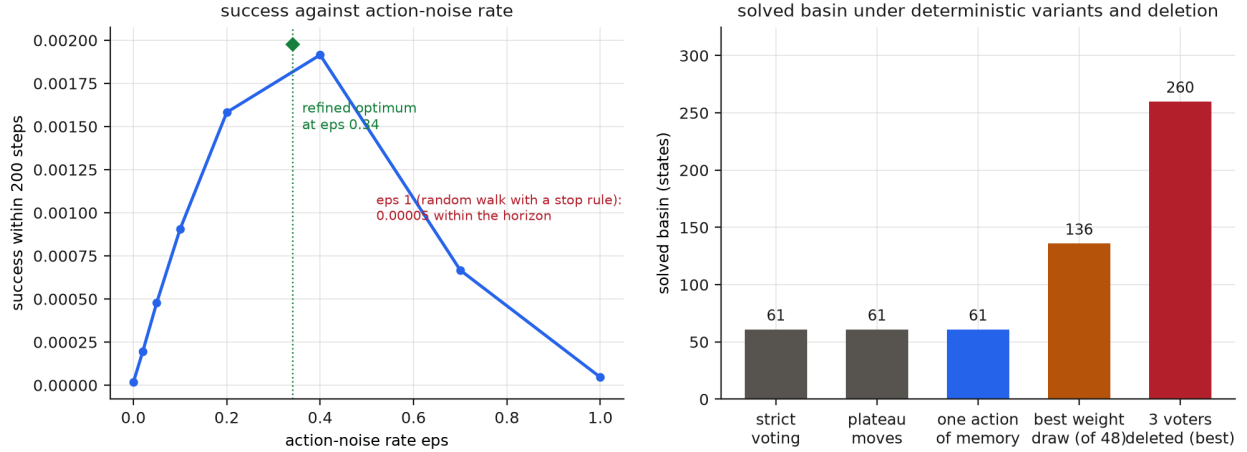


Figure 2: Repairs. Left: success within 200 steps against action-noise rate, with the refined interior optimum marked; the endpoint at $\epsilon = 1$ is the random-walk control. Right: solved basin under the deterministic variants, all at 61, and under weighting and deletion, the only interventions that move it.

Faults as probes

We now use faults as the method intends, to reveal structure.

Deleting voters shows that the intact collective was over-coordinated. Removing 1 of the 7 agents raises the mean basin over the 7 deletion patterns to 62.9; removing 2 raises it to 69.8; removing 3 raises it to 91.8, with the best 3-deletion pattern reaching 260, more than 4 times the intact basin, and the worst falling to 19 (Figure 2, right). Reduced participation of components thus increases system-level competence here, and the exhaustive analysis identifies the mechanism: fewer votes make the score landscape coarser, fewer strict-improvement requirements mean fewer premature stops, and for some patterns the surviving agents' greedy steps happen to lead toward solved. The effect is a matter of tie-breaking, and the same fault class that quadruples the basin also reduces it by two thirds, depending on which agents fail. Under the noise channel the direction reverses: success at $\varepsilon = 0.1$ falls monotonically from 0.000905 intact to 0.000333 with 3 agents deleted, so the redundant votes helped the stochastic explorer while obstructing the deterministic climber. The same fault class gives opposite results depending on whether the other repair channel is active.

Deleting moves shows the boundary between incompetence and impossibility. Removing one face's 3 generators leaves a subgroup whose orbit through solved contains 29,160 states, 0.79 percent of the space. From the other 99.2 percent no controller can recover, because the fixed corner's neighbour on the deleted face can no longer move. Removing 2 faces leaves an orbit of 4 states. This algebraic fact, computed as the orbit of the surviving generators, distinguishes failure attributable to the controller from failure imposed by the dynamics, which no behavioral experiment alone can do (Figure 3, left).

Conflicting targets show how arbitration resolves normative conflict. We give m agents the solved target and $7 - m$ agents a target B at distance 6, and keep the strict arbiter. The summed vote is then a mixed potential, the dynamics ascend it, and cycles are impossible, as the exhaustive classification confirms at every split. The enumeration adds the scale of the resulting landscape: between 728,698 and 777,584 equilibria at every split, nearly all of them compromises that neither faction encodes. The count is largest at the most balanced split of 4 against 3, where the two factions' basins are 55 and 53. Both encoded targets remain equilibria at every ratio, so the conflict is resolved by basin sizes and never by elimination of a target, and balance increases the number of equilibria (Figure 3, right). Softening the arbiter to allow lateral moves restores 128,244 cycles of length up to 4. Each outcome is a recurrent class of competing update rules, predicted in kind though not in number by the potential structure, and none requires a negotiating collective.

Retargeting shows a factorization and its limit. The controller is target-relative: assigning every agent a new pose from any reachable configuration leaves the machinery unchanged. Group translation makes the new closed loop conjugate to the old, so the basin of the new target equals that of the old, and the computation gives 61 for target B, 61 for target C and 61 for solved. The separation of competency mechanism from target specification, which Levin's framework emphasizes in morphogenesis, holds here completely and by construction (Levin, 2022; Fields and Levin, 2022), and provides no competence. A mechanism can be fully reprogrammable and uniformly weak, so reprogrammability and competence are independent properties, and evidence for one is not evidence for the other.

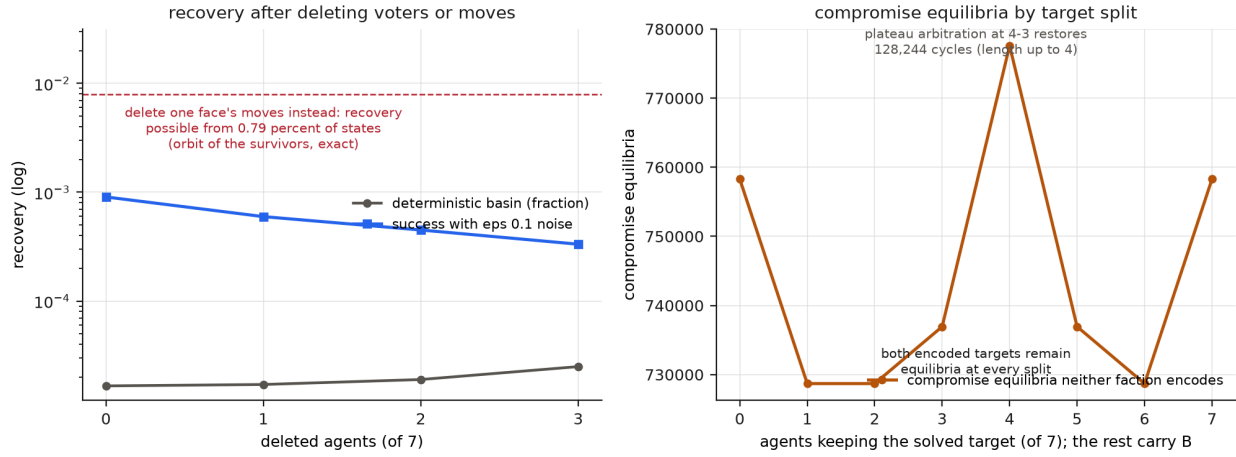


Figure 3: Faults as probes. Left: deleting voters raises the deterministic basin while lowering noisy success, and deleting one face's moves makes recovery algebraically impossible from 99.2 percent of states. Right: compromise equilibria by target split under strict arbitration, with both encoded targets remaining equilibria at every ratio.

A ladder of goal-directedness

Claims about goal-directedness come in grades. G0 is optimization: actions statistically improve an objective. G1 is stabilization: broad convergence and closure, Dijkstra's property. G2 is robust regulation: recovery from perturbation and component faults, homeostasis in the cybernetic sense that begins with purpose as feedback (Rosenblueth, Wiener and Bigelow, 1943) and includes the regulator theorem, by which every good regulator is a model of what it regulates (Ashby, 1956; Conant and Ashby, 1970). G3 is flexible control: multiple routes, context-sensitive action and useful temporary regression. G4 is reprogrammable competence: the same machinery redirected across targets and unseen faults without redesign. G5 is autonomous normativity: a system that constitutes and maintains its own conditions of existence, the enactive criterion of individuality, normativity and asymmetry, which no software cube meets (Barandiaran, Di Paolo and Rohde, 2009). The controllers above sit at G0, reach G1 on only 0.0017 percent of the space, and exhibit the G4 factorization in a form that shows it to be independent of competence. Goal language is licensed where it predicts, as in the intentional stance (Dennett, 1987), and the intermediate vocabulary of control mechanisms often does the same work with fewer metaphysical commitments (Bich and Bechtel, 2022).

An emergent collective competency requires more than that no individual can solve the task, which is easy to arrange. It requires that no component and not the arbiter contains the global policy, that the interaction is causally necessary, that the collective generalizes to targets and faults absent from its construction, and that a macro-level goal model predicts the outcomes of interventions more compactly than the micro-level description. The cube generates exact instances of what does not count: a random walk with an absorbing target, a distance oracle translated into votes, a target-aware arbiter, improvement from ordinary annealing, a stable limit cycle described as a preference, robustness from duplicated votes, and success that depends on a benevolent scheduler. Nothing

here meets the strong condition. The exhaustive substrate instead yields the complete list of deflationary explanations that a stronger claim on any substrate must exclude item by item.

Conditions for faultization as a method

The neighbouring traditions are close: fault injection validates recovery machinery (Hsueh, Tsai and Iyer, 1997), chaos engineering stresses deployed services (Basiri et al., 2016), lesion and knock-out studies map function to components, and self-stabilization already covers recovery without a handler. Faultization adds something only under 7 conditions drawn from the failures the cube exposed. Scale must be an experimental variable, with the decomposition varied. Fault classes must be causally separated; deletion, structural and normative faults gave 3 different kinds of result here. The arbiter must be audited, since one relaxation of arbitration reorganized a fifth of the state space. Matched search controls must be run, since the random walk with a stop rule converges universally and the annealing curve accounts for beneficial noise. Novelty must be tested on held-out cases. A macro-level goal description must improve compression and prediction. And faults must be read as probes, asking what structure becomes inferable because of the fault, as when the two opposite effects of deletion located the redundancy, or when the loss of generators separated the controller's failure from the group's constraint.

Under these conditions, ordinary fault tolerance asks whether a specified service persists, and faultization asks what the fault reveals about the causal and descriptive scale of the service, including whether a different service, attractor or agent has appeared.

Objections

The first objection is that every result was foreseeable: greedy local voting fails on a coupled combinatorial group, noise anneals, deletion perturbs tie-breaking, and potentials cannot cycle. The initial hypotheses of this study, however, predicted that memory would enlarge the basin and that noise would rescue collectives with deleted agents, and both were wrong in the measured direction; the effect of deletion ran opposite to the effect of noise; and exact sizes such as 61, 758,286, 145,953 and 29,160 do not follow from qualitative argument. The second objection is that the substrate is unfavourable to emergence, with rigid pieces, fixed topology, no growth, no metabolism and an externally specified target. This is a scope condition: the cube tests the deflationary catalogue, and a phenomenon that survives it warrants testing on richer substrates, where the same accounting is harder and more necessary. The third objection is that the deflationary results pre-judge Levin's programme. They apply that programme's own methodological recommendations (barrier-conditioned controls, matched stochastic baselines, and the authors' statement that nothing operates beyond the rules) on a substrate where they can be carried to completion, and they raise the evidential bar without reaching a verdict about tissues. The fourth objection is that the strict arbiter is unrealistic, since real collectives are asynchronous and noisy. The plateau, memory, noise and heterogeneity variants address that objection exhaustively, and the basin moved from 61 only when the collective was damaged or randomized, never when it was enriched.

Falsification

The numerical claims are computed exhaustively, so falsification applies to the framework. The method adds nothing if its 7 conditions add nothing to existing fault-injection practice, which would appear as studies meeting them reaching the same conclusions as studies that do not. The deflationary catalogue fails as a standard if some system meets it trivially, with no component holding the global policy, an audited arbiter, held-out generalization and a predictive macro-model arising without design effort, in which case the strong condition must be strengthened. The substrate claim fails if the lessons do not transfer: if controllers that pass these tests are helpless on the $3 \times 3 \times 3$ cube or on substrates with growth and changing topology, the tests measured properties of the cube. Specific findings are also open to direct challenge: a memory design that enlarges the basin, a repair channel under which deletion helps both the deterministic climber and the stochastic explorer, or an arbitration scheme under which an encoded target is eliminated would each overturn a result reported here.

Conclusion

The pocket cube is not an organism, and its 3,674,160 states allowed every claim to be counted. Local correctness, granted in its most complete form, composed into almost no competence, and the group's parity explains why the final joint step cannot be seen locally. Memory, lateral moves and heterogeneity left the basin unchanged or moved it only through tie-breaking, and the one repair that helped was ordinary exploration, priced against a random walk that eventually solves every state and almost never within the horizon. Damage was the only intervention that enlarged the deterministic basin, for reasons the enumeration identifies, and deletion of generators separated what a controller failed to do from what the group made impossible. A fully reprogrammable mechanism carried its 61-state basin to every target. Where robust, flexible and reprogrammable goal-directedness is claimed for a collective, whether of arrays, automata or tissues, these results provide an enumerated catalogue of cheaper explanations that the claim must exclude.

Reproducibility

The simulation, its 3 figures and every model number quoted above regenerate from one command in the repository (`simulation/run_all.py`). The first run builds the full transition tables in a few minutes of pure Python; each experiment then runs exhaustively in numpy over all 3,674,160 states, deterministically except for the seeded weight draws, and identically on rerun. 28 invariants fail the run if broken, among them the exact reproduction of the published distance shells, the absence of states with 6 correct corners, the equality of the strict, plateau and memory basins at 61, the interior noise optimum against both endpoint controls and its golden-section refinement, the monotone rise of the deterministic deletion basins against the monotone fall of the noisy ones, the exact orbit of the surviving generators, the absence of cycles under strict arbitration with conflicting targets, and the exact conjugacy of retargeted basins. The statistics from the self-sorting array experiments enter `results.json` as cited records with their source.

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